Biochemistry Cume **December 3, 2010**

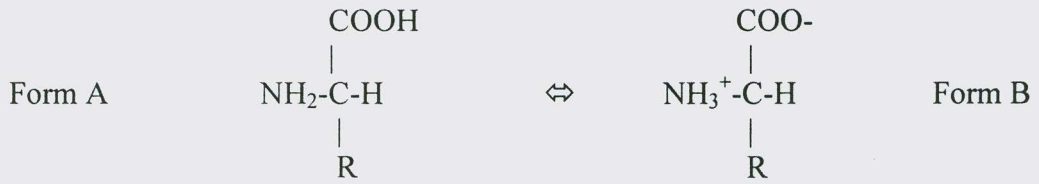
1. (40 points) For the sequencing of proteins, it is often desirable to have a reagent react with the deprotonated N-terminal amino group (pKa = 8) but not with the amino group of lysine R-groups (pKa = 10.5). Under what experimental conditions might this be accomplished? Use calculations to justify your answer.

2. (60 points) Consider the equilibrium for the dissociation of a weak acid

$$HA \Leftrightarrow A^- + H^+$$

a. Show that for such a reaction, $\Delta G^o = 2.3\ RT\ pKa$.

Now, consider the following reaction (for an amino acid in which R contains no ionizable group):



b. What is the relationship between ΔG^o and the two pKa values associated with this reaction?

c. Although form A is often considered as being absent, it is actually present in a very small amount. What is the ratio of (B)/(A)? (Assume that the carboxyl pKa is 2.3 and that the amino pKa is 9.6.)